

Deployment Runbook — Family Music Scrobbler PWA

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A complete operational guide for deploying, maintaining, and verifying the Family Music Scrobbler PWA on the Intel NUC (Fedora Server) with Synology DS223 backup integration.

1. Deployment Environment Overview

Hardware

- Intel NUC (Core i5-4250U, 8GB DDR3L RAM)
- 120GB mSATA SSD
- Synology DS223 NAS (backup target)

Operating System

- Fedora Server (latest stable)

Core Services

- Docker
 - Docker Compose
 - PostgreSQL
 - FastAPI backend
 - React PWA frontend
 - APScheduler ingestion worker
-

2. Directory Layout on Production Server

```
/opt/family-scrobbler/  
  backend/  
  frontend/  
  infra/  
  docker/  
  scripts/  
  secrets/
```

Secrets Directory

```
/opt/family-scrobber/secrets/  
  refresh_token_key  
  vapid_private_key  
  vapid_public_key
```

Secrets are mounted via Docker secrets.

3. First-Time Deployment Steps

Step 1 - Install Docker & Docker Compose

```
sudo dnf install docker-ce docker-ce-cli containerd.io  
sudo systemctl enable --now docker
```

Step 2 - Clone the Repository

```
cd /opt  
sudo git clone https://github.com/Notascooby25/family-scrobber.git  
cd family-scrobber
```

Step 3 - Create Production .env

Populate environment variables for:

- PostgreSQL
- Spotify OAuth
- AES encryption key
- VAPID keys

Step 4 - Load Secrets

Place secrets into /opt/family-scrobber/secrets/.

Step 5 - Build & Start the Stack

```
docker compose -f infra/docker-compose.yml up --build -d
```

Step 6 - Verify Services

Check running containers:

```
docker ps
```

Check backend health:

```
curl http://localhost:8000/healthz
```

4. Updating the Deployment

Step 1 - Pull Latest Code

```
cd /opt/family-scrobber  
sudo git pull
```

Step 2 - Rebuild Containers

```
docker compose -f infra/docker-compose.yml up --build -d
```

Step 3 - Run Database Migrations

```
docker exec backend alembic upgrade head
```

Step 4 - Verify Worker Status

Check logs:

```
docker logs backend --tail 50
```

5. Database Backup Procedure

Backups run via scripts/verify_backup.sh.

Backup Steps

1. Perform pg_dump of the live database.
2. Sync dump to Synology DS223 via rsync.
3. Spin up a temporary PostgreSQL container.
4. Restore the dump.
5. Run integrity checks.

Manual Execution

```
cd /opt/family-scrobber/scripts  
sudo bash verify_backup.sh
```

Cron Scheduling Example

```
0 */6 * * * /opt/family-scrobber/scripts/verify_backup.sh
```

Runs every 6 hours.

6. Monitoring & Observability

Health Check Endpoint

```
GET /healthz
```

Returns JSON with:

- DB connectivity
- Worker heartbeat
- API status

Logs

All services output structured JSON logs.

View logs:

```
docker logs backend
```

7. Security Practices

- All refresh tokens encrypted using AES key stored in Docker secrets.
 - VAPID keys stored in secrets directory.
 - No secrets committed to Git.
 - Production .env stored only on the NUC.
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8. Maintenance Tasks

Restarting Services

```
docker compose -f infra/docker-compose.yml restart
```

Clearing Old Images

```
docker system prune -af
```

Checking Disk Usage

```
df -h
```

9. Disaster Recovery

Step 1 - Retrieve Latest Backup

Located on Synology DS223.

Step 2 - Restore Database

```
docker exec -i postgres psql -U postgres < latest_dump.sql
```

Step 3 - Rebuild Stack

```
docker compose -f infra/docker-compose.yml up --build -d
```

Step 4 - Validate Health

```
curl http://localhost:8000/healthz
```

10. Deployment Checklist

- ☐ Secrets loaded
- ☐ Production .env configured
- ☐ Docker stack running
- ☐ /healthz returns OK
- ☐ Worker logs show polling
- ☐ Backups verified
- ☐ PWA loads on mobile + desktop

This runbook provides everything required to deploy, maintain, and recover the Family Music Scrobbler PWA in your home-lab environment. Ready for operational use.